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Computer Science & Engineering

Experiment 6

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Subject Name: Database Management System **Subject Code: 24CSH-298**

AIM: This experiment focuses on understanding and implementing cursors in PL/SQL to process multiple rows returned by a query. It demonstrates the use of implicit cursors, explicit cursors, and cursor attributes to apply business logic on database records in a row-by-row manner.

OBJECTIVES:

- To understand the concept of cursors in PL/SQL
- To differentiate between implicit and explicit cursors
- To use cursor attributes such as %FOUND, %NOTFOUND, %ROWCOUNT, and %ISOPEN
- To process multiple rows using explicit cursors
- To apply business logic using cursor-based processing

SOFTWARE REQUIREMENTS:

- Oracle Database Express Edition (Oracle XE)
- Oracle SQL Developer

PROCEDURE/STEPS:

1. Start the system and log in
2. Open Oracle SQL Developer

3. Connect to the required database
4. Open the SQL Worksheet / Query Tool
5. Create the employee table
6. Insert sample data into the table
7. Execute PL/SQL blocks demonstrating cursor usage
8. Observe outputs in DBMS Output panel
9. Save the work and take screenshots

PRACTICAL/EXPERIMENTAL STEPS:

Table Creation

```
CREATE TABLE employees (
   emp_id NUMBER PRIMARY KEY,
   emp_name VARCHAR2(50),
   salary NUMBER,
   department VARCHAR2(50)
);
INSERT INTO employees VALUES (101, 'Amit', 30000, 'HR');
INSERT INTO employees VALUES (102, 'Neha', 45000, 'IT');
INSERT INTO employees VALUES (103, 'Raj', 50000, 'Finance');
INSERT INTO employees VALUES (104, 'Simran', 40000, 'IT');
INSERT INTO employees VALUES (105, 'Karan', 35000, 'HR');

COMMIT;
```

Implicit Cursor Example

```
SET SERVEROUTPUT ON;

DECLARE
   v_rows NUMBER;
BEGIN
   UPDATE employees
   SET salary = salary + 5000
```

```
   WHERE emp_id = 101;

   v_rows := SQL%ROWCOUNT;

   DBMS_OUTPUT.PUT_LINE('Rows Updated: ' || v_rows);
END;
/
```

Explicit Cursor Example

```
DECLARE
   CURSOR emp_cursor IS
      SELECT emp_id, emp_name, salary FROM employees;

   v_id employees.emp_id%TYPE;
   v_name employees.emp_name%TYPE;
   v_salary employees.salary%TYPE;

BEGIN
   OPEN emp_cursor;

   LOOP
      FETCH emp_cursor INTO v_id, v_name, v_salary;
      EXIT WHEN emp_cursor%NOTFOUND;

      DBMS_OUTPUT.PUT_LINE(
         'ID: ' || v_id ||
         ', Name: ' || v_name ||
         ', Salary: ' || v_salary
      );
   END LOOP;

   CLOSE emp_cursor;
END;
/
```

Cursor Attributes Example

```
DECLARE
   CURSOR emp_cursor IS
      SELECT emp_name, salary FROM employees;

   v_name employees.emp_name%TYPE;
   v_salary employees.salary%TYPE;

BEGIN
   OPEN emp_cursor;

   LOOP
      FETCH emp_cursor INTO v_name, v_salary;
      EXIT WHEN emp_cursor%NOTFOUND;

      DBMS_OUTPUT.PUT_LINE(
         'Row ' || emp_cursor%ROWCOUNT ||
         ': ' || v_name || ' earns ' || v_salary
      );
   END LOOP;

   IF emp_cursor%ISOPEN THEN
      DBMS_OUTPUT.PUT_LINE('Cursor is open, now closing...');
      CLOSE emp_cursor;
   END IF;
END;
/
```

Business Logic Example

```
DECLARE
   CURSOR emp_cursor IS
      SELECT emp_id, salary FROM employees;
```

```
    v_id employees.emp_id%TYPE;
    v_salary employees.salary%TYPE;

BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id, v_salary;
        EXIT WHEN emp_cursor%NOTFOUND;

        IF v_salary < 40000 THEN
            UPDATE employees
            SET salary = salary + 2000
            WHERE emp_id = v_id;

            DBMS_OUTPUT.PUT_LINE(
                'Bonus added to Employee ID: ' || v_id
            );
        END IF;
    END LOOP;

    CLOSE emp_cursor;
    COMMIT;
END;
/
```

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OUTPUT:(Screenshots)

Implicit Cursor Example

[SQL Worksheet]*

```
INSERT INTO employees VALUES (105, 'Kartik', 35000, 'HR');

COMMIT;
DECLARE
    v_rows NUMBER;
BEGIN
    UPDATE employees
    SET salary = salary + 5000
    WHERE emp_id = 101;

    -- Implicit cursor attribute
    v_rows := SQL%ROWCOUNT;

    DBMS_OUTPUT.PUT_LINE('Rows Updated: ' || v_rows);
END;
/
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> DECLARE
       v_rows NUMBER;
     BEGIN
       UPDATE employees...
Show more...

Rows Updated: 1

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008
```

Explicit Cursor Example

[SQL Worksheet]*

```
BEGIN
    OPEN emp_cursor;

    LOOP
        FETCH emp_cursor INTO v_id, v_name, v_salary;

        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            'ID: ' || v_id ||
            ', Name: ' || v_name ||
            ', Salary: ' || v_salary
        );
    END LOOP;
    CLOSE emp_cursor;
END;
/
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Show more...

ID: 101, Name: Yuvraj, Salary: 30000
ID: 102, Name: Jaskaran, Salary: 45000
ID: 103, Name: Rohit, Salary: 50000
ID: 104, Name: Simran, Salary: 40000
ID: 105, Name: Kartik, Salary: 35000

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.100
```

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Cursor Attributes Example

[SQL Worksheet]*

```
    LOOP
        FETCH emp_cursor INTO v_name, v_salary;

        EXIT WHEN emp_cursor%NOTFOUND;

        DBMS_OUTPUT.PUT_LINE(
            'Row ' || emp_cursor%ROWCOUNT ||
            ': ' || v_name || ' earns ' || v_salary
        );
    END LOOP;

    IF emp_cursor%ISOPEN THEN
        DBMS_OUTPUT.PUT_LINE('Cursor is open, now closing...');
        CLOSE emp_cursor;
    END IF;
END;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Row 1: Yuvraj earns 30000
Row 2: Jaskaran earns 45000
Row 3: Rohit earns 50000
Row 4: Simran earns 40000
Row 5: Kartik earns 35000
Cursor is open, now closing...

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.107
```

Business Logic Example

[SQL Worksheet]*

```
        EXIT WHEN emp_cursor%NOTFOUND;

        IF v_salary < 40000 THEN
            UPDATE employees
            SET salary = salary + 2000
            WHERE emp_id = v_id;

            DBMS_OUTPUT.PUT_LINE(
                'Bonus added to Employee ID: ' || v_id
            );
        END IF;
    END LOOP;

    CLOSE emp_cursor;
    COMMIT;
END;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
        CURSOR emp_cursor IS
            SELECT emp_id, salary FROM employees;
    ...
Show more...

Bonus added to Employee ID: 101
Bonus added to Employee ID: 105

PL/SQL procedure successfully completed.

Elapsed: 00:00:00.014
```

Final Table Output

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[SQL Worksheet]*

```
        IF v_salary < 40000 THEN
            UPDATE employees
            SET salary = salary + 2000
            WHERE emp_id = v_id;

            DBMS_OUTPUT.PUT_LINE(
                'Bonus added to Employee ID: ' || v_id
            );
        END IF;
    END LOOP;

    CLOSE emp_cursor;
    COMMIT;
END;
/
SELECT * FROM employees;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

Download Execution time: 0.006 seconds

	EMP_ID	EMP_NAME	SALARY	DEPARTMENT
1	101	Yuvraj	37000	HR
2	102	Jaskaran	45000	IT
3	103	Rohit	50000	Finance
4	104	Simran	40000	IT
5	105	Kartik	37000	HR

I/O ANALYSIS:

Input:

- Employee ID
- Employee Name
- Salary
- Department

Output:

- Updated employee salary
- Row-by-row processed employee records
- Salary-based business logic execution

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LEARNING OUTCOMES:

- Learned how cursors work in PL/SQL
- Understood implicit and explicit cursors
- Used cursor attributes effectively
- Implemented row-by-row data processing
- Applied real-world business logic